

① How many solutions has the problem?

✓ a) $x'' + t^2 x = 0, x(0) = 0$

b) $x'' + t^2 x = 0, x(0) = 0, x'(0) = 0$

c) $x'' + t^2 x = 0, x(0) = 0, x'(0) = 0, x''(0) = 1$

b) $\rightarrow$ IVP.

Obs. that $x=0$ sol. of b) $\stackrel{\exists! \text{ Theorem}}{\text{}} \stackrel{(\text{guess it.})}{\text{}} x=0$ the unique solution.

c) (H1): $x=0$ the unique function that verifies the ^{first 2} conditions of c). Since this function does not verify the last condition $\Rightarrow$ c) has no sol.

(H2): Assume c) has a sol. $\varphi: \mathbb{R} \rightarrow \mathbb{R}, \varphi''(t) + t^2 \varphi(t) = 0, \forall t \in \mathbb{R}.$

$\varphi(0) = 0, \varphi'(0) = 0, \varphi''(0) = 1$
Check for $t=0 \Rightarrow \varphi''(0) + 0 \cdot \varphi(0) = 0 \Rightarrow 1 + 0 = 0$ ctr.

a) not an IVP $\rightarrow$ make it an IVP $\rightarrow$ add condition $x'(0) = \eta, \eta \in \mathbb{R}.$

This IVP has a unique sol. $\varphi_\eta: \mathbb{R} \rightarrow \mathbb{R} \Rightarrow$ family of solutions, $\forall \eta \in \mathbb{R}.$

② Let $v \in C^1(\mathbb{R})$ with $v(0) = 0, v'(0) \neq 0, x v(x) > 0, \forall x \in \mathbb{R}^*.$ Planar system:

$$\begin{cases} \dot{x} = y \\ \dot{y} = -v(x) \end{cases}$$

a) f.i. in $\mathbb{R}^2$?

b) stability of $(0,0)$ equiv?

$$\begin{aligned} \text{a) } \frac{\dot{y}}{\dot{x}} &= \frac{dy}{dx} \Rightarrow \frac{-v(x)}{y} = \frac{dy}{dx} \Rightarrow -v(x) dx = y dy \quad \bigg| \int \\ &\Rightarrow \int_0^x -v(s) ds = \int_0^y \frac{1}{2} y dy \\ &\Rightarrow -\int_0^x v(s) ds = \frac{1}{2} y^2 + C \\ &\Rightarrow \underbrace{\frac{1}{2} y^2 + \int_0^x v(s) ds}_H(x,y) = C \end{aligned}$$

$$H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x,y) = \frac{1}{2} y^2 + \int_0^x v(s) ds.$$

$$\nabla H(x,y) = (v(x), y)$$

check: $\frac{\partial H}{\partial x}(x,y) \cdot \dot{x} + \frac{\partial H}{\partial y}(x,y) \cdot \dot{y} \stackrel{?}{=} 0$

$$\Rightarrow y \cdot v(x) + y \cdot v(x) = 0, \forall (x,y)$$

$$\nabla H(x,y) = 0 \Leftrightarrow \begin{cases} v(x) = 0 \\ y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \Rightarrow (0,0) \text{ the only critical point of } H. \\ \Rightarrow H \text{ not locally ct.}$$

b) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{bmatrix} y \\ -v(x) \end{bmatrix}$, $f(0,0) = (0,0) \Rightarrow (0,0)$ equil.

$$Jf(x,y) = \begin{bmatrix} 0 & 1 \\ -v'(x) & 0 \end{bmatrix}$$

$$Jf(0,0) = \begin{bmatrix} 0 & 1 \\ -v'(0) & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & 1 \\ -v'(0) & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + v'(0) = 0 \Rightarrow \lambda_{1,2} = \pm i \sqrt{v'(0)} \Rightarrow \text{CENTER} \\ \Rightarrow \text{not hyp. because } \operatorname{Re}(\lambda) = 0.$$

~~$v'(x) = x$~~ $x v(x) > 0, (\forall) x \in (-\infty, 0) \Rightarrow v(x) < 0, (\forall) x \in (-\infty, 0) \quad \left. \begin{array}{l} ? \\ \end{array} \right\} \Rightarrow$
 $x v(x) < 0, (\forall) x \in (0, \infty) \Rightarrow v(x) < 0, (\forall) x \in (0, \infty)$
 $\Rightarrow \exists v \dots \Rightarrow v$ has the same sign as x (*)

$$\Rightarrow \lambda_{\mathbb{R}} = \pm i \sqrt{v'(0)}$$

Linearization fails for non-hyperbolic equil. $\Rightarrow$ use H .

$$H(x,y) = \frac{1}{2}y^2 + \int_0^x v(s) ds = \frac{1}{2}y^2 + \underbrace{\dots}_{>0} \underbrace{\dots}_{>0} > 0 \Rightarrow H(x,y) > 0, (\forall) (x,y) \neq (0,0) \\ H(0,0) = 0$$

$$\int_0^x v(s) ds$$

$$x > 0 \Rightarrow v(s) > 0, x \neq 0$$

$\Downarrow$
 $\begin{cases} H \text{ has a strict local min. at } (0,0) \\ H \text{ ct. along trajectories} \end{cases} \Rightarrow (0,0) \text{ stable. (not a. stable because } H \text{ is conserved)}$

LINEARIZATION FAILS

3. Prey-Predator: $\begin{cases} \dot{x} = x(1-y) \\ \dot{y} = -y(2-x) \end{cases}$ a) f.i.? check.
 b) (2,1) equil? hyp?

a) $\dots H(x,y) = y + x - 2xy - 2\ln x = C$

? $H: (0,\infty) \rightarrow \mathbb{R}$

$$\nabla H(x,y) = \left(\frac{\partial H}{\partial x}, \frac{\partial H}{\partial y} \right) = \left(1 - \frac{2}{x}, 1 - \frac{1}{y} \right)$$

$$\frac{\partial H}{\partial x}(x,y) \cdot \dot{x} + \frac{\partial H}{\partial y}(x,y) \cdot \dot{y} = 0, (\forall) (x,y) \in (0,\infty) \times (0,\infty)$$

$\dots \Rightarrow$ yes, f.i.

$\nabla H(x,y) = (0,0) \Rightarrow (2,1)$ unique critical point $\Rightarrow$ not locally ct. H

$$f(x,y) = \begin{bmatrix} x - xy \\ -2y + xy \end{bmatrix}$$

$$f(2,1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (2,1) \text{ equil. point}$$

$$Jf(x,y) = \begin{bmatrix} 1-y & -x \\ y & -2+xy \end{bmatrix}$$

$$Jf(2,1) = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + 2 = 0 \Rightarrow \lambda = \pm i\sqrt{2}$$

$$\operatorname{Re}(\lambda) = 0 \Rightarrow (2,1) \text{ not hyp.}$$

g Non-linear planar system: $\dot{x} = -y + x \cdot (1 - x^2 - y^2)$
 (we) $\dot{y} = x + y \cdot (1 - x^2 - y^2)$
 ...

a) Check $\ell(t, 1, 0) = (\cos t, \sin t)$, $\forall t \in \mathbb{R}$.

$\ell(t, 1, 0)$ = the sol. of the system that verifies the initial conditions.

$$x(0) = 1, y(0) = 0$$

$$\text{Let } x(t) = \cos t, t \in \mathbb{R}.$$

$$y(t) = \sin t$$

$$\text{We have to check: } \begin{cases} x'(t) = -y(t) + x(t) \cdot (1 - x^2(t) - y^2(t)) \\ y'(t) = x(t) + y(t) \cdot (1 - x^2(t) - y^2(t)) \\ x(0) = 1 \\ y(0) = 0 \end{cases}, t \in \mathbb{R}$$

$$\begin{cases} x'(t) = -y(t) \Rightarrow -\sin t = -\sin t \checkmark \\ y'(t) = x(t) \Rightarrow \cos t = \cos t \\ x(0) = 1 \Rightarrow \cos 0 = 1 \text{ true} \\ y(0) = 0 \Rightarrow \sin 0 = 0 \text{ true} \end{cases} \Rightarrow \ell(t, 1, 0) \text{ indeed sol.}$$

b) Phase portrait? using polar coord.

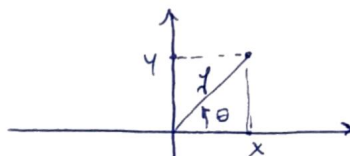
/c)

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \quad \left| \begin{aligned} \Rightarrow x^2 + y^2 &= r^2 (\cos^2 \theta + \sin^2 \theta) = r^2 \\ \tan \theta &= \frac{y}{x} \end{aligned} \right.$$

$$r^2 = x^2 + y^2 \quad |' \Rightarrow 2 \cdot r \cdot \dot{r} = 2 \cdot x \cdot \dot{x} + 2 \cdot y \cdot \dot{y} \Rightarrow r \dot{r}^2 = x \dot{x} + y \dot{y}$$

$$\tan \theta = \frac{y}{x} \quad |' \Rightarrow (1 + \tan^2 \theta) \cdot \dot{\theta} = \frac{\dot{y} \cdot x - x \cdot \dot{y}}{x^2}$$

$$\begin{cases} \ddot{x} = x\dot{x} + y\dot{y} & (1) \\ (1 + \tan^2 \theta) \cdot \dot{\theta} = \frac{\dot{y}x - x\dot{y}}{x^2} & (2) \end{cases}$$



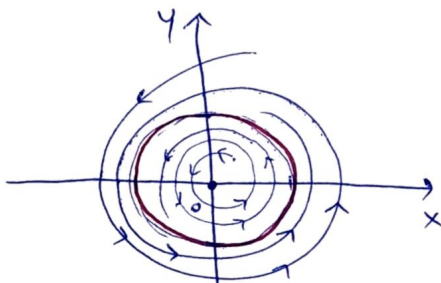
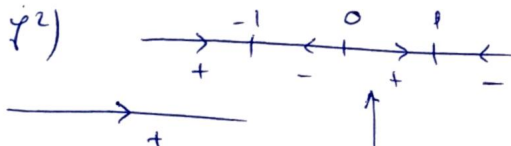
b)

$$(1): \ddot{r} = x(\dots) + y(\dots) = (1 - x^2 - y^2)(x^2 + y^2) = (1 - r^2)r^2 \quad | : r \neq 0 \\ \Rightarrow \ddot{r} = r(1 - r^2)$$

$$\boxed{1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}} \quad , \quad r = r \cos \theta$$

$$(2): \frac{\dot{\theta}}{\cos^2 \theta} = \frac{r^2}{r^2 \cos^2 \theta} \Rightarrow \dot{\theta} = 1 > 0 \Rightarrow \text{each orbit rotates around the origin counterclockwise. (+ trigon.)} \\ \frac{\dot{y}x - x\dot{y}}{x^2} = \frac{\dots}{x^2} \quad r = 1 \Rightarrow x^2 + y^2 = 1$$

The new system:
$$\begin{cases} \ddot{r} = r(1 - r^2) \\ \dot{\theta} = 1 \end{cases}$$



c) Reading the phase portrait $\rightarrow (0,0)$ repeller? $\exists$ attractor?

iii) $D: C^1(\mathbb{R}) \rightarrow C(\mathbb{R}), D(x) = x', L(x) = (x+2)^3$

$L: C^3(\mathbb{R}) \rightarrow C(\mathbb{R}), L = L(D) = (D+2)^3$

$L(x) = (D+2)^3(x) \quad L(x) = (D^3(x) + 2)^3 = (x' + 2)^3$

$\rightarrow L(x) = x''' + 6x'' + 12x' + 8x$

a) $D(e^{-t}) = (e^{-t})' = -e^{-t}$

$(D+2)(e^{-t}) = (e^{-t})''' = (e^{-t})'' = (e^{-t})' = -e^{-t} \\ = D(e^{-t}) + 2 \cdot (e^{-t}) = -e^{-t} + 2 \cdot e^{-t} = e^{-t}$

$L(e^{-t}) = (D+2)^3(e^{-t}) = e^{-t}$

$(D+2)(e^{-t}) = e^{-t}$

$(D+2)(e^{-t}) = e^{-t}$

$(D+2)(e^{-t}) = e^{-t}$

~~$D(x) = x'$~~
 ~~$(D+1)(x) = x^4$~~
 ~~$(D+2)(x) = x^4$~~
 ~~$D(x) = x'$~~

$(D+1)(x) = D(x) + x$

$(D+2)(x) = D(x) + 2x$

(4)

$$b) D(e^{-2t}) = (e^{-2t})' = -2e^{-2t}$$

$$(D+2)(e^{-2t}) = \dots$$

$$c) L = L(D) = (D+2)^3$$

$$(D+2)^3(x) = 0$$

$$x_h = (c_1 + c_2 t + c_3 t^2) e^{-2t}$$

$$a) \Rightarrow \frac{L(x) = A \cdot e^{-t}}{L(e^{-t}) = e^{-t}}, A > 0$$

$$x_p = A \cdot e^{-t}$$

$$x(t) = (c_1 + c_2 t + c_3 t^2) e^{-2t} + A e^{-t}$$

$$d) L(x) = A e^{-2t}, A > 0.$$

$$L(t^3 e^{-2t}) = 6 e^{-2t}$$

$$x_p = \frac{A}{6} t^3 e^{-2t}$$

$$L(x_p) = A e^{-2t}$$

$$x(t) = (c_1 + c_2 t + c_3 t^2) e^{-2t} + \frac{A}{6} e^{-2t} \cdot t^3$$

Barbălat Lemma

$$f: \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \text{ fct. s.t. } \exists \lim_{t \rightarrow \infty} f(t), \lim_{t \rightarrow \infty} f'(t), \text{ finite } \xrightarrow{?} \lim_{t \rightarrow \infty} f'(t) = 0.$$

$$\text{Mean Value Theorem: } \exists \xi_m \in [m, m+1] \text{ s.t. } \frac{f(m+1) - f(m)}{m+1 - m} = f'(\xi_m)$$

$$\Rightarrow f(m+1) - f(m) = \cancel{f'} f'(\xi_m)$$

$$\exists \lim_{t \rightarrow \infty} f(t) \text{ finite} \Rightarrow \lim_{m \rightarrow \infty} f(m) = L$$

$$\lim_{m \rightarrow \infty} f(m+1) = L$$

$$\underline{\hspace{10em}} (-)$$

$$\Rightarrow \lim_{m \rightarrow \infty} f(m+1) - \lim_{m \rightarrow \infty} f(m) = L - L = 0$$

$$\Rightarrow \lim_{m \rightarrow \infty} f'(\xi_m) = 0$$

$$\left. \begin{array}{l} \lim_{m \rightarrow \infty} \xi_m = \infty \\ \exists \lim_{t \rightarrow \infty} f'(t) = L' \end{array} \right\} \Rightarrow \lim_{m \rightarrow \infty} f'(\xi_m) = L' = 0 \Rightarrow \lim_{t \rightarrow \infty} f'(t) = 0.$$

③ $f: \mathbb{R} \rightarrow \mathbb{R}$, c' fct, $f: \mathbb{R} \rightarrow \mathbb{R}$ sol. of $\dot{x} = f(x)$ s.t. $\lim_{t \rightarrow \infty} \varphi(t) = 2$
 ? 2 equil of $\dot{x} = f(x)$.

$$\left. \begin{array}{l} \varphi \text{ sol. of } \dot{x} = f(x) \Rightarrow \varphi'(t) = f(\varphi(t)), (\forall) t \in \mathbb{R} \\ \lim_{t \rightarrow \infty} \varphi(t) = 2 \in \mathbb{R} \\ \text{if cont.} \end{array} \right\} \Rightarrow \lim_{t \rightarrow \infty} f(\varphi(t)) = f(2) \Rightarrow \lim_{t \rightarrow \infty} \varphi'(t) = f(2)$$

$$\left. \begin{array}{l} \lim_{t \rightarrow \infty} \varphi'(t) = f(2) \text{ finite} \\ \lim_{t \rightarrow \infty} \varphi(t) = 2 \text{ finite} \end{array} \right\} \xrightarrow{\text{Barbalat Lemma}} \lim_{t \rightarrow \infty} \varphi'(t) = 0 \Rightarrow f(2) = 0 \Rightarrow 2 \text{ equil. point.}$$

④ $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $H: \mathbb{R}^2 \rightarrow \mathbb{R} \rightarrow$ two c' fct. s.t. H is a f.i. in $\mathbb{R}^2$ of the planar system $\dot{x} = f(x)$. $\dot{x} = f(x)$ no global attractor?

(*) $H \text{ f.i.} \Rightarrow H(\varphi(t, \eta)) = H(\eta), (\forall) \eta \in \mathbb{R}^2.$

Assume $p =$ global attractor.

$$\Rightarrow \lim_{t \rightarrow \infty} \varphi(t, \eta) = p, (\forall) \eta \in \mathbb{R}^2. \left. \begin{array}{l} H \text{ cont.} \end{array} \right\} \Rightarrow \lim_{t \rightarrow \infty} H(\varphi(t, \eta)) = H(p) \Rightarrow H(\varphi(t, \eta)) = H(\eta) *$$

-) $\lim_{t \rightarrow \infty} H(\eta) = H(p) \Rightarrow \eta = p. H(\eta) = H(p), (\forall) \eta \in \mathbb{R}^2.$

$\Rightarrow H$ is ct. on $\mathbb{R}^2$
 $H \text{ f.i. not locally ct} \Rightarrow \text{ctr.}$

$\Rightarrow \dot{x} = f(x)$ no global attr.

⑤ $w, \mu > 0$ fixed parameters.

$\theta'' + \mu \theta' + w^2 \sin \theta = 0 \rightarrow$ the second order nonlinear equation
 $\theta(t) = 0, (\forall) t \in \mathbb{R} \rightarrow$ attractor?

let $x = \theta, y = \theta'$
 $\Rightarrow \begin{cases} x' = \theta' = y \\ y' = \theta'' = -\mu y - w^2 \sin \theta \end{cases}$

$$A = \begin{pmatrix} 0 & y \\ -w^2 x & -\mu y \end{pmatrix}$$

$\theta(t) = 0 \rightarrow (x, y) = 0$

$$J = \begin{pmatrix} 0 & 1 \\ -w^2 & -\mu \end{pmatrix}$$

$\Rightarrow \begin{vmatrix} -\lambda & 1 \\ -w^2 & -\mu - \lambda \end{vmatrix} = 0 \Rightarrow -\lambda(-\mu - \lambda) + w^2 = 0$

$\Rightarrow \lambda^2 + \mu \lambda + w^2 = 0 \Rightarrow \Delta = \mu^2 - 4 \cdot \lambda^2 \cdot w^2 < 0$

$\Rightarrow \lambda_{1,2} = \dots \Rightarrow \text{Re}(\lambda) < 0 \Rightarrow \text{STABLE}$
 HYP $\Rightarrow$ equil. has same stability

⑥

$$\omega > 0.$$

$$\begin{cases} \theta'' + \omega^2 \theta = 0 \\ \theta(0) = \frac{\pi}{6} \\ \theta'(0) = 0 \end{cases}$$

• oscillations?

• after how much time $\rightarrow$ init. state? ✓

• phase portrait? $\rightarrow$ planar dyn. sys. $\rightarrow$ type? ✓

• $\theta(0) = 0$ stable, but not attractor.

$$\lambda^2 + \omega^2 = 0 \Rightarrow \lambda = \pm i\omega.$$

$$\theta(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$\theta(0) = c_1 + 0 = \frac{\pi}{6}$$

$$\theta'(t) = -\omega \cdot c_1 \sin(\omega t) + \omega \cdot c_2 \cos(\omega t)$$

$$\theta'(0) = \underbrace{\omega \cdot c_2}_{>0} = 0 \Rightarrow c_2 = 0$$

$$\Rightarrow \theta(t) = \frac{\pi}{6} \cos(\omega t).$$

• periodic if $\exists T > 0$ s.t. $f(t+T) = f(t)$

$$\cos(\phi + 2\pi) = \cos \phi \quad \Rightarrow \text{in our case } \phi = \omega t$$

$$\sin(\phi + 2\pi) = \sin \phi \quad \Rightarrow \omega \cdot (t+T) \Rightarrow \omega t + \omega T = \omega t + 2k\pi$$

$$\Rightarrow \omega T = 2k\pi$$

$$\Rightarrow T = \frac{2k\pi}{\omega}$$

$$T_{\min} = \frac{2\pi}{\omega} \quad (k=1)$$

$$\text{Initial state: } \begin{cases} \theta(t) = \frac{\pi}{6} \\ \theta'(t) = 0 \end{cases} \Rightarrow \begin{cases} \frac{\pi}{6} \cos \omega t = \frac{\pi}{6} \Rightarrow \cos \omega t = 1 \\ -\frac{\pi}{6} \omega \sin \omega t = 0 \Rightarrow \sin \omega t = 0 \end{cases}$$

$$\Rightarrow \omega t = 2k\pi \Rightarrow t = \frac{2k\pi}{\omega} \quad \Rightarrow \text{pendulum returns to init. state after } \frac{2\pi}{\omega} \text{ min.}$$

Associated planar system:

$$x = \theta, y = \theta'$$

$$\begin{cases} x' = \theta' = y \\ y' = -\omega^2 x \end{cases} \Rightarrow A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}, \quad Jf(x,y) = \begin{bmatrix} 0 & 1 \\ -\omega^2 & 0 \end{bmatrix}$$

$$f(x,y) = \begin{bmatrix} y \\ -\omega^2 x \end{bmatrix} \Rightarrow \begin{vmatrix} -\lambda & 1 \\ -\omega^2 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + \omega^2 = 0 \Rightarrow \lambda = \pm i\omega.$$

$$\operatorname{Re}(\lambda) = 0 \Rightarrow (0,0) \text{ center.}$$

$$f(x,y) = 0 \Leftrightarrow \begin{bmatrix} y \\ -\omega^2 x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Leftrightarrow x = y = 0$$

Phase portrait:

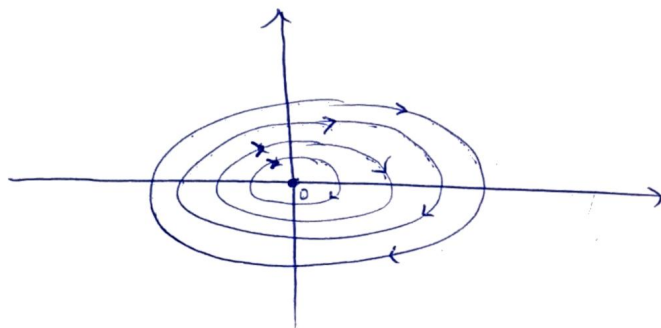
$$\frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} \Rightarrow \frac{-w^2 x}{y} = \frac{dy}{dx} \Rightarrow$$

$$\Rightarrow -w^2 x dx = y dy \quad | \int \Rightarrow -w^2 \cdot \int x dx = \int y dy \Rightarrow -w^2 \cdot \frac{x^2}{2} = \frac{y^2}{2} + C$$

$$\Rightarrow \underbrace{-\frac{w^2 x^2}{2} - \frac{y^2}{2}}_H = C \quad | \cdot (-1) \Rightarrow \frac{w^2 x^2}{2} + \frac{y^2}{2} = C.$$

$$H(x,y) = \frac{y^2}{2} + \frac{w^2 x^2}{2} = C. \Rightarrow \text{ellipses around } (0,0).$$

$$(x,y) = (1,0) \Rightarrow \begin{cases} \dot{x} = y = 0 \\ \dot{y} = -w^2 < 0 \end{cases} \Rightarrow \text{clockwise direction.}$$



$\operatorname{Re}(\lambda) = 0 \Rightarrow$ linearization method fails. $\Rightarrow$ use H .

$$H(x,y) = \underbrace{\frac{w^2 x^2}{2}}_{\geq 0} + \underbrace{\frac{y^2}{2}}_{\geq 0} \geq 0, \quad H \text{ ct. along trajectories} \Rightarrow (0,0) \text{ stable.}$$

$$H(x,y) = 0 \Leftrightarrow (x,y) = (0,0) \Rightarrow (0,0) \text{ strict min. of } H.$$

$(0,0)$ NOT ATTRACTOR (but stable) because $\neq 0$ trajectories are close ellipses, so they don't converge to $(0,0)$.

$w > 0; \quad \theta'' + w^2 \sin \theta = 0$

a) equil? lin. method.

$$\text{let } x = \theta, \quad y = \theta'$$

$$\begin{cases} x' = \theta' = y \\ y' = \theta'' = -w^2 \sin x \end{cases}$$

$$\text{let } f(x,y) = \begin{bmatrix} y \\ -w^2 \sin x \end{bmatrix}, \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$f(x,y) = 0 \Leftrightarrow \begin{cases} y = 0 \\ -w^2 \sin x = 0 \Rightarrow \sin x = 0 \Rightarrow x = 2k\pi, \quad k \in \mathbb{Z}. \end{cases}$$

$$\Rightarrow \text{equil. } (k\pi, 0)$$

$$Jf(x,y) = \begin{bmatrix} 0 & 1 \\ -w^2 \cos x & 0 \end{bmatrix}$$

$$Jf(k\bar{u}, 0) = \begin{bmatrix} 0 & 1 \\ -w^2(-1)^k & 0 \end{bmatrix} \Rightarrow \begin{vmatrix} -\lambda & 1 \\ -w^2(-1)^k & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + w^2(-1)^k = 0$$

$$\cos k\bar{u} = (-1)^k$$

• Case I: $k = 2s$ (even)

$$\Rightarrow \lambda^2 + w^2 = 0 \Rightarrow \lambda^2 = -w^2 \Rightarrow \lambda = \pm iw$$

$\text{Re}(\lambda) = 0 \Rightarrow$ non-hyp $\Rightarrow$ lin. fails.

• Case II: $k = 2s+1$ (odd)

$$\Rightarrow \lambda^2 - w^2 = 0 \Rightarrow \lambda^2 = w^2 \Rightarrow \lambda = \pm w$$

$\lambda_1 < 0 < \lambda_2 \Rightarrow$ saddle, unstable.

b)

$$\text{for case I: } H(x,y) \Rightarrow \frac{\dot{y}}{\dot{x}} = \frac{dy}{dx}$$

$$\Rightarrow \frac{-w^2 \sin x}{y} = \frac{dy}{dx}$$

$$\Rightarrow -w^2 \sin x dx = y dy \quad \int$$

$$\Rightarrow -w^2 \int \sin x dx = \int y dy \Rightarrow -w^2 \cdot (-\cos x) = \frac{y^2}{2} + C \Rightarrow$$

$$\Rightarrow w^2 \cdot \cos x = \frac{y^2}{2} + C \Rightarrow w^2 \cdot \cos x - \frac{y^2}{2} = C \quad | + w^2$$

$$H(x,y)$$

$$-1 \leq \cos x \leq 1 \quad | \cdot w^2 > 0$$

$$-w^2 \leq w^2 \cdot \cos x \leq w^2 \quad | - \frac{y^2}{2}$$

$$-w^2 - \frac{y^2}{2} \leq w^2 \cdot \cos x - \frac{y^2}{2} \leq w^2 - \frac{y^2}{2}$$

$$1 - \cos x \geq 0$$

$$H(0,0) = 0$$

$$H(x,y) \geq 0$$

$$\frac{\partial H}{\partial x}(x,y) \dot{x} + \frac{\partial H}{\partial y}(x,y) \dot{y} \stackrel{?}{=} 0$$

$$\Rightarrow \cancel{y \cdot \dot{x} + \dot{y} \cdot x}$$

$$\Rightarrow -w^2 \sin x \cdot y + (-y) \cdot (-w^2 \sin x) = 0 \text{ true. } \Rightarrow H \text{ global f.i.}$$

$$H(0,0) = 0$$

NOT ATTR. (0,0) bc. $\neq 0$ sol. don't converge to (0,0)

$H(x,y) > 0$; $H(x,y) = 0 \Leftrightarrow (x,y) = (0,0) \Rightarrow H$ has a strict local min at (0,0).

$\Rightarrow H$ cf. along trajectories $\Rightarrow$ nearby sol $\in$ nearby lev. curves $\Rightarrow$ (0,0) stable

$$\Rightarrow H(x,y) = \frac{y^2}{2} - w^2 \cos x = C$$

$$\Rightarrow \frac{y^2}{2} + w^2(1 - \cos x) = H(x,y) = C$$

$$-\frac{24}{2}$$

$$\frac{\partial H}{\partial x} = -w^2 \sin x$$

$$\frac{\partial H}{\partial y} = -y$$

8. $\begin{cases} \dot{x} = -y - x(x^2 + y^2) \\ \dot{y} = x - y(x^2 + y^2) \end{cases}$ non-linear planar system.

a) (0,0) non-hyp. eq. u.e.

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, f(x,y) = \begin{bmatrix} -y - x^3 - xy^2 \\ x - yx^2 - y^3 \end{bmatrix}$$

$$Jf(x,y) = \begin{bmatrix} -3x^2 - y^2 & -1 - 2xy \\ 1 - 2yx & -x^2 - 3y^2 \end{bmatrix}$$

$$Jf(0,0) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Rightarrow \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i.$$

$$\operatorname{Re}(\lambda) = 0$$

b) phase portrait (polar coord).

$\Rightarrow$ (0,0) non-hyp.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2 \Rightarrow 2x\dot{x} + 2y\dot{y} = 2r\dot{r} \Rightarrow x\dot{x} + y\dot{y} = r\dot{r}$$

$$\left| \frac{y}{x} = \tan \theta \right|'$$

$$\Rightarrow x(-y - x(x^2 + y^2)) + y(x - y(x^2 + y^2)) = r\dot{r}$$

$$\Rightarrow -xy - x^2r^2 + xy - y^2r^2 = r\dot{r}$$

$$\Rightarrow r^2(-x^2 - y^2) = r\dot{r} \quad | : r \neq 0$$

$$\Rightarrow \cancel{r} \Rightarrow r(-r^2) = \dot{r}$$

$$\Rightarrow \dot{r} = -r^3$$

$$\tan \theta = \frac{y}{x} \Rightarrow (1 + \tan^2 \theta) \dot{\theta} = \frac{\dot{y}x - y\dot{x}}{x^2}$$

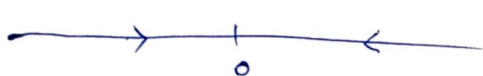
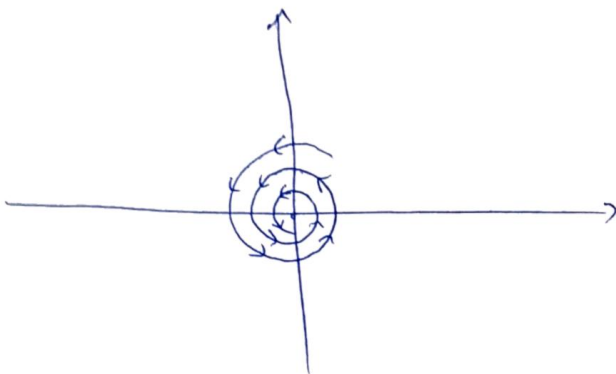
$$\Rightarrow \frac{\dot{\theta}}{\cos^2 \theta} = \frac{\dot{y}x - y\dot{x}}{x^2} \Rightarrow \frac{\dot{\theta}}{\cos^2 \theta} = \frac{\dot{y}x - y\dot{x}}{r^2 \cos^2 \theta}$$

$$r^2 \cos^2 \theta = x^2$$

$$\begin{aligned} \dot{y}x - y\dot{x} &= (x - y(x^2 + y^2))x - y(-y - x(x^2 + y^2)) \\ &= x^2 + y^2 = r^2 \end{aligned}$$

$$\Rightarrow \frac{\dot{\theta}}{\cos^2 \theta} = \frac{r^2}{r^2 \cos^2 \theta} \Rightarrow \dot{\theta} = 1 > 0 \Rightarrow \text{counterclockwise (trigon.)}$$

$\Rightarrow$ New system: $\begin{cases} \dot{r} = -r^3 < 0 \text{ for } r > 0 \Rightarrow \text{radius decreases.} \\ \dot{\theta} = 1 \Rightarrow \text{clock.} \end{cases}$



$\ddot{f}$



$\dot{\theta}$

c) (0,0) equiv. attr.? (because radius decreases).

$$\dot{f} = -f^3 < 0$$

$$\frac{df}{dt} = -f^3 \Rightarrow \frac{df}{f^3} = -dt \Rightarrow -\frac{1}{2f^2} = -t + C \Rightarrow$$

$$\Rightarrow \frac{1}{f^2} = 2t + C$$

$f(t) \rightarrow 0$ as $t \rightarrow \infty$. $\Rightarrow$ (0,0) attractor.

10) $\begin{cases} \dot{x} = -x + 2y \\ \dot{y} = 2x - 4y \end{cases}$ lin. planar system

a) reduction method?

$$\ddot{x} = -\dot{x} + 2\dot{y} = -\dot{x} + 2(2x - 4y) = -\dot{x} + 4x - 8y$$

$$\dot{x} = -x + 2y \Rightarrow \dot{x} + x = 2y \Rightarrow \frac{\dot{x} + x}{2} = y$$

$$\Rightarrow \ddot{x} = -\dot{x} + 4x - 8 \cdot \frac{\dot{x} + x}{2} = -\dot{x} + 4x - 4\dot{x} - 4x = -5\dot{x}$$

$$\Rightarrow \ddot{x} + 5\dot{x} = 0$$

$$x^2 + 5x = 0 \Rightarrow x(x+5) = 0 \Rightarrow x \in \{0, -5\}$$

$$x=0 \rightarrow e^{0 \cdot t}$$

$$\Rightarrow x(t) = c_1 \cdot e^{0 \cdot t} + c_2 \cdot e^{-5t} = c_1 + c_2 \cdot e^{-5t}$$

$$x=-5 \rightarrow e^{-5t}$$

$$x'(t) = -5c_2 \cdot e^{-5t}$$

$$y(t) = \frac{-5c_2 \cdot e^{-5t} + c_1 + c_2 \cdot e^{-5t}}{2} = \frac{c_1 - 4c_2 \cdot e^{-5t}}{2}$$

b) char. eq. method.

$$A = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \quad \det(A - \lambda I_2) = 0$$

$$\Rightarrow \begin{vmatrix} -1-\lambda & 2 \\ 2 & -4-\lambda \end{vmatrix} = 0 \Rightarrow (-1-\lambda)(-4-\lambda) - 4 = 0 \Rightarrow$$

$$\Rightarrow (1+\lambda)(4+\lambda) - 4 = 0 \Rightarrow \lambda^2 + 5\lambda + 4 = 0 \Rightarrow \lambda(\lambda+5) = 0.$$

$$\lambda_1 = 0$$

$$\lambda_2 = -5$$

$$\lambda_2 < 0 \leq \lambda_1$$

$$\lambda_2 < \lambda_1 \leq 0$$

$\Rightarrow$ NODE.

$$\lambda_1 = 0 \Rightarrow \det(A - 0 \cdot I_2) \overset{v_1}{=} 0$$

$\Rightarrow$ stable.

$$\Rightarrow Av = 0 \Rightarrow \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \begin{cases} -x + 2y = 0 \\ 2x - 4y = 0 \end{cases} \mid :2 \Rightarrow \begin{cases} -x + 2y = 0 \\ x - 2y = 0 \end{cases}$$

$$\Rightarrow v_1 = \begin{bmatrix} 2y \\ y \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow e^{\lambda_1 t} \cdot v_1 \Rightarrow x = 2y$$

$$\lambda_2 = -5 \Rightarrow (A + 5I_2)v_2 = 0$$

$$= e^{0 \cdot t} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} \rightarrow x = 2y$$

$$\Rightarrow \left(\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \begin{cases} 4x + 2y = 0 \\ 2x + y = 0 \end{cases} \Rightarrow \begin{cases} 2x + y = 0 \\ 2x + y = 0 \end{cases}$$

$$\Rightarrow y = -2x \Rightarrow v_2 = \begin{bmatrix} x \\ -2x \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \rightarrow e^{-5t} \cdot \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$X(t) = c_1 \cdot e^{0 \cdot t} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \cdot e^{-5t} \cdot \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$X(t) = c_1 \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \cdot e^{-5t} \cdot \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$X = \begin{pmatrix} 2c_1 \\ c_1 \end{pmatrix} + \begin{pmatrix} c_2 \cdot e^{-5t} \\ -2c_2 \cdot e^{-5t} \end{pmatrix}$$

$$\Rightarrow \begin{cases} x(t) = 2c_1 + c_2 \cdot e^{-5t} \\ y(t) = c_1 - 2c_2 \cdot e^{-5t} \end{cases}$$

$$\dot{x} = -x + xy = x(y-1)$$

$$\dot{y} = -2y + 3y^2 = y(3y-2)$$

a) equil., stability

$$(0,0), (0, \frac{2}{3})$$

$$J(x,y) = \begin{pmatrix} y-1 & x \\ 0 & -2+6y \end{pmatrix}$$

$$J(0,0) = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \Rightarrow \lambda_1 = -1, \lambda_2 = -2 \Rightarrow \text{attractor, node.}$$

$$J(0, \frac{2}{3}) = \begin{pmatrix} -\frac{1}{3} & 0 \\ 0 & 2 \end{pmatrix} \Rightarrow \lambda_1 = -\frac{1}{3}, \lambda_2 = 2 \Rightarrow \text{saddle, unstable.}$$

b) $\varphi(t, 0, \frac{2}{3}), \varphi(t, 4, 0), \varphi(t, 1, \frac{2}{3})$

$(0, \frac{2}{3})$ equil. $\Rightarrow \varphi(t, 0, \frac{2}{3}) = (0, \frac{2}{3})$

both $\rightarrow 0$ equil for $\dot{x}, \dot{y}$ equil. for $\dot{y}$.

$y(0)=0 \Rightarrow y(t)=0 \Rightarrow \dot{x}(t) = -x(t) + 0 = -x(t); \dot{y}(0)=0.$

$x(0)=4 \Rightarrow x(t) = 4e^{-t} \rightarrow x(0)=4 \text{ STOP.}$

$\rightarrow y(0)=0 \Rightarrow y(t)=0$

~~$(4 \text{ not equil. value})$~~

~~$$\dot{x} = x(y-1)$$~~

$$\dot{x} = -x \Rightarrow \dot{x} + x = 0$$

$$x + 1 = 0 \Rightarrow x = -1$$

$$\Rightarrow \varphi(t, 4, 0) = (4e^{-t}, 0).$$

$$\Rightarrow x(t) = c_1 \cdot e^{-t} = 4 \cdot e^{-t}.$$

$$x(0)=4 \uparrow$$

$\varphi(t, 1, \frac{2}{3}) \rightarrow x(0)=1 \text{ STOP.}$

$$\rightarrow y(0)=\frac{2}{3} \Rightarrow y(t)=\frac{2}{3}$$

~~$$x(0)=1 \Rightarrow x(t)=1 \quad (1 \text{ not equil. value for } \dot{x} = x(y-1))$$~~

$$y(0)=\frac{2}{3} \Rightarrow y(t)=\frac{2}{3} \quad (\text{equil value for } \dot{y} = -2y + 3y^2)$$

$$\dot{x} = x(\frac{2}{3}-1) = -\frac{1}{3}x \Rightarrow \dot{x} + \frac{1}{3}x = 0$$

$$x + \frac{1}{3} = 0$$

$$\Rightarrow x = -\frac{1}{3}.$$

$$\Rightarrow \varphi(t, 1, \frac{2}{3}) = (e^{-\frac{1}{3}t}, \frac{2}{3})$$

c). $(0, \frac{2}{3}) : \Rightarrow \Gamma = \{(0, \frac{2}{3})\}$ orbit. $\Rightarrow e \cdot e^{-\frac{1}{3}t} = x(t) = e^{-\frac{1}{3}t}.$

$(4, 0) : \Rightarrow \Gamma = \{(x, 0) : x > 0\} \quad x(0)=1 \Rightarrow e=1$

$(1, \frac{2}{3}) : \Rightarrow \Gamma = \{(x, \frac{2}{3}) : x > 0\}$